How skilful are AI-based precipitation hindcasts in the 2023 Indian summer monsoon?

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Motivation

- The recent development of global AI Weather Prediction (AIWP) models has shown potential to revolutionize weather forecasting.
- Although many studies have evaluated AIWP output, their precipitation skill remains relatively underexplored.
- A case in point is forecasting Indian summer monsoon (ISM) precipitation which remains challenging due to substantial spatial and temporal variability.
- In this study, we verify the precipitation hindcast skill of AIWP models during the 2023 ISM and compare their performance to a physics-based numerical model.

Data

- The AIWP models used are AIFS (Moldovan et al. 2026), GraphCast (Lam et al. 2023) and FuXi (Chen et al. 2023). Their performance is compared against the NWP model IFS.
- Model performance is evaluated using daily accumulated precipitation (verified against IMERG) and daily mean winds (verified against ERA5).

Methods

Verification Metric	What it Quantifies	Interpretation & Range
Root-Mean-Squared Error (RMSE)	Overall magnitude of the difference between forecast and observed precipitation	0 = perfect; larger values → larger absolute error (units: mm day^{-1})
Pattern Correlation Coefficient (PCC)	Linear association between forecast and observations in space	–1 = perfect inverse, 0 = no linear association, +1 = perfect match
SAL-S (Structure error)	Tendency of a precipitating object to appear smoother than observations (Wernli et al; 2008)	Sharper < 0 < Smoother; –2 → 0 → +2
SAL-A (Amplitude error)	Mean-intensity bias (too wet/dry)	Drier < 0 < Wetter; –2 → 0 → +2
SAL-L (Location error)	Displacement and spread of precipitating objects	0 = features perfectly positioned; 2 = maximal mismatch
Stable Equitable Error in Probability Space (SEEPS)	How well the forecast hits climatology-based dry / light / heavy rain categories (Rodwell et al; 2010)	0 = perfect hit; values > 0 indicate error

Table 1. Verification metrics used in this study. Each captures a different aspect of precipitation skill: error magnitude, pattern, structure, amplitude, location and rain categorization.

Results

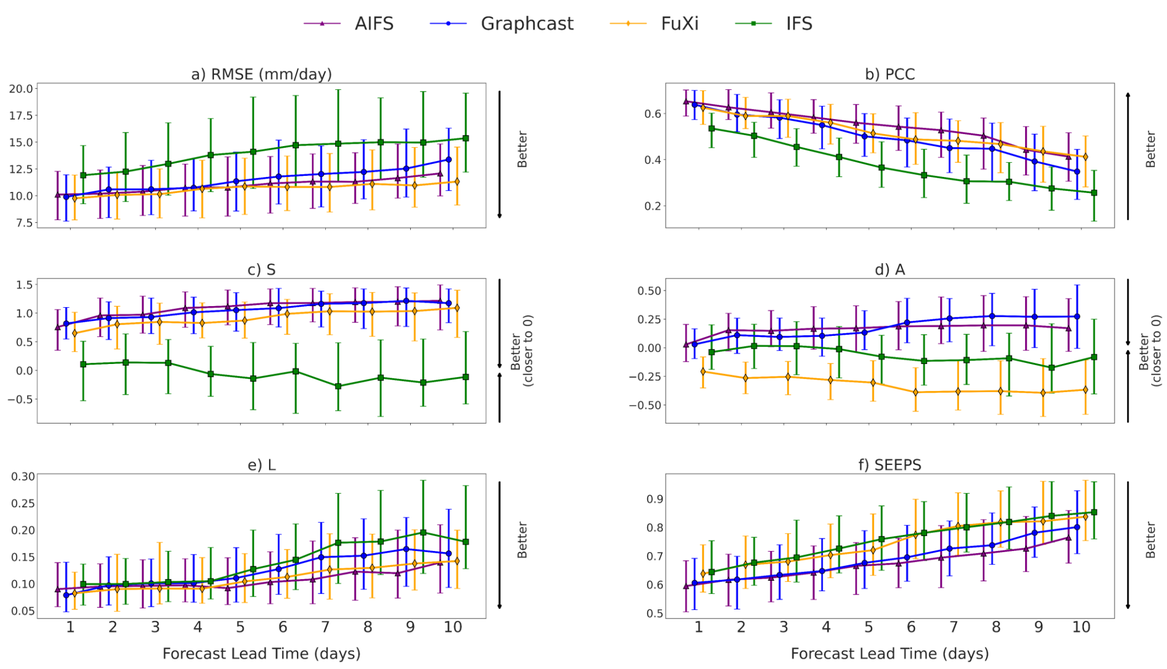


Fig 1. Precipitation verification metrics over India for each model during JJAS 2023 as a function of lead time. Panels show (a) RMSE, (b) PCC, (c) SAL-Structure error (S), (d) SAL-Amplitude error (A), (e) SAL-Location error (L) and (f) SEEPS for AIFS (purple), GraphCast (blue), FuXi (orange) and IFS (green). The x-axis gives hindcast lead time (lead 7 means each day's hindcast was initialized 7 days earlier). Shapes mark the median of the hindcast scores across the season; whiskers span the inter-quartile range.

Verification Metric	Lead 1	Lead 5	Lead 10
RMSE	-	-	FuXi
PCC	AIFS	AIFS	AIFS
S	IFS	IFS	IFS
A	-	-	-
L	-	-	-
SEEPS	AIFS	AIFS	AIFS

Table 2. Model that significantly outperforms all others for each metric at lead days 1, 5 and 10 (95% bootstrap confidence intervals of paired differences). A dash means no model was significantly better.

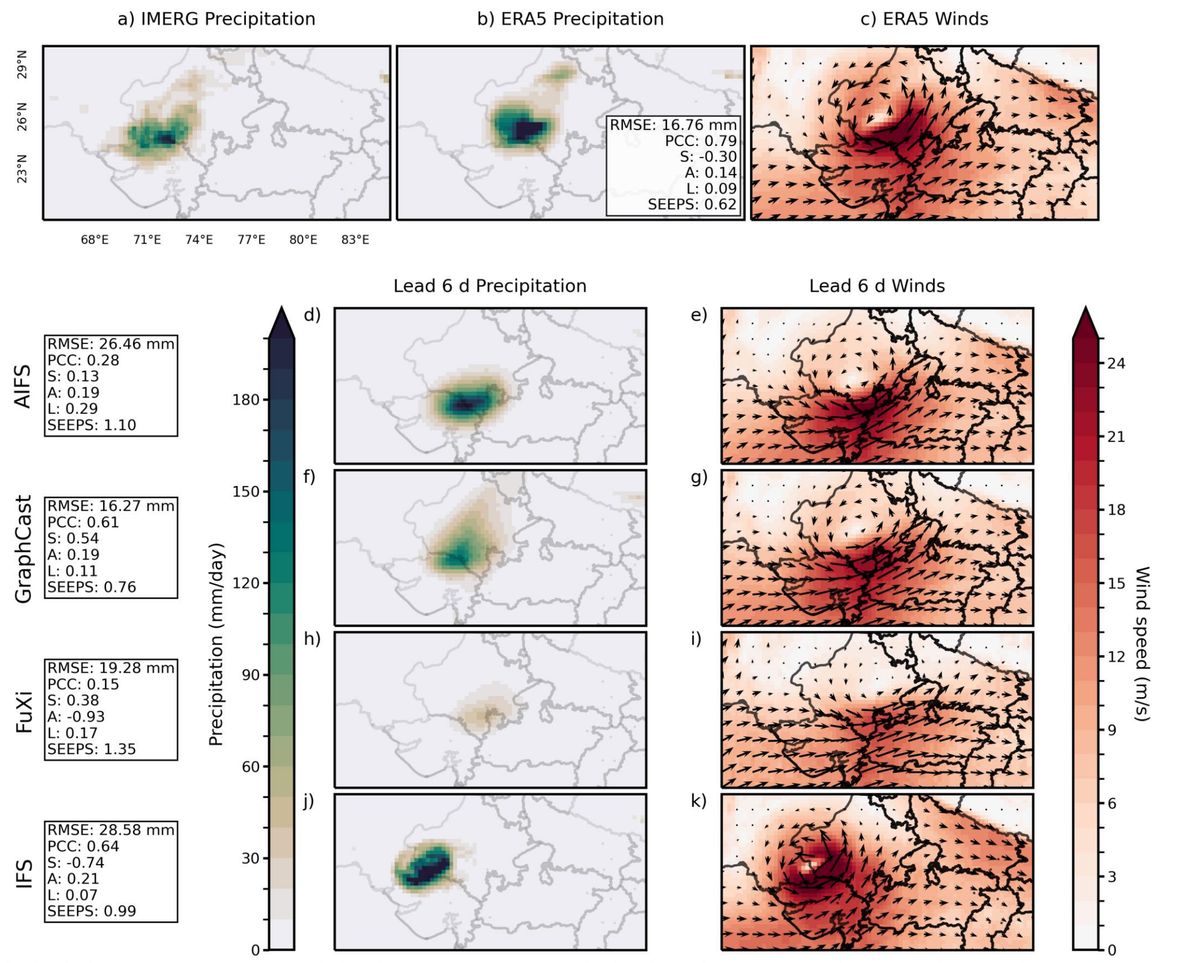


Fig 2. Daily accumulated precipitation (mm/day) and 850 hPa daily mean horizontal winds valid on 17 June 2023 during tropical cyclone 'Biparjoy'. Top row: a) IMERG observation, b) ERA5 precipitation and c) ERA5 winds. Rows 2–4: Precipitation verification scores corresponding to each model against IMERG: RMSE, PCC, SAL components (S, A, L) and SEEPS (left column), Precipitation (middle column) and winds (right column) of AIFS (d,e), GraphCast (f,g), FuXi (h,i) and IFS (j,k) hindcasts initialized 6 days before valid time.

Key findings

- AIWP models exhibited better performance in terms of RMSE and PCC than IFS (Fig 1a, Fig 1b, Table 2).
- AIWP models performed poorly in terms of structure error showing only positive values, indicating overly–smoothed precipitation structures (Fig 1c, Fig 2d,f,h, Table 2).
- FuXi consistently underpredicts extreme rainfall, showing a conditional bias (Fig 2h).

Conclusions

- AIWP models generally outperform IFS in RMSE and PCC, but show large positive structure error, indicating overly–smoothed precipitation.
- AIFS emerged as the overall best–performing model as it outperforms other models in RMSE, PCC and SEEPS across most lead times.
- These findings also show that model skill is metric–dependent and choosing between an AIWP or a numerical model should hinge on the forecaster's priority.
- This varying performance of precipitation of AIWP models using different metrics could foster further research in the need for training precipitation on better and more suitable loss functions in future AIWP models.

References

1. Moldovan, G., Pinnington, E., Prieto Nemesio, A., Lang, S., Ben Bouallègue, Z., Dramsch, J., Alexe, M., Santa Cruz, M., Hahner, S., Cook, H., Theissen, H., Clare, M., O'Brien, C., Polster, J., Magnusson, L., Mertes, G., Pinault, F., Raoult, B., de Rosnay, P., Forbes, R., and Chantry, M.: AIFS Single 1.1.0: an update to ECMWF's machine-learned weather forecast model AIFS, Geosci. Model Dev., 19, 4703–4724, https://doi.org/10.5194/gmd-19-4703-2026, 2026.
2. Remi Lam et al., Learning skillful medium-range global weather forecasting.Science382,1416-1421(2023).DOI:10.1126/science.adi2336
3. Chen, L., Zhong, X., Zhang, F. et al. FuXi: a cascade machine learning forecasting system for 15-day global weather forecast. npj Clim Atmos Sci 6, 190 (2023). https://doi.org/10.1038/s41612-023-00512-1.
4. Wernli, H., M. Paulat, M. Hagen, and C. Frei, 2008: SAL—A Novel Quality Measure for the Verification of Quantitative Precipitation Forecasts. *Mon. Weather Rev.*, **136**, 4470–4487, https://doi.org/10.1175/2008MWR2415.1.
5. Rodwell, M.J., Richardson, D.S., Hewson, T.D. and Haiden, T. (2010), A new equitable score suitable for verifying precipitation in numerical weather prediction. Q.J.R. Meteorol. Soc., 136: 1344-1363. https://doi.org/10.1002/qj.656

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